

Auto Ticket Classification using Flow Designer

TEAM ID	DATE	DOCUMENT TYPE	PLATFORM
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Project Team Members

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1. INTRODUCTION

1.1 Project Overview

The **Auto Ticket Classification using Flow Designer** project addresses a core operational inefficiency in a school IT helpdesk: the manual reviewing of incoming incident requests and manually assigning appropriate categories. The helpdesk receives continuous requests from students and teachers regarding various technical issues, including Wi-Fi connectivity, projector hardware failures, password resets, and performance degradation (slow computers). The implemented solution uses ServiceNow Flow Designer to evaluate keywords in an incident's *Short Description* and *Description* fields, automatically classifying the ticket immediately upon creation.

1.2 Purpose & Objectives

The primary purpose of this project is to eliminate repetitive manual classification, accelerate ticket processing and routing, and establish a maintainable no-code automation layer configured directly via ServiceNow logic rules.

Core Objectives

- Automatically classify incidents upon creation.
- Eliminate manual routing lag for IT agents.
- Improve overall incident resolution speed.

Key Architectural Gains

- Zero-code, low-maintenance configuration.
- Consistent rule-based category matching.
- Direct integration with downstream support.

2. IDEATION PHASE

2.1 Problem Statement

School IT staff currently spend significant daily hours manually reviewing incoming support requests to assign categories before assignment groups can take action. This manual overhead creates bottlenecks, introduces inconsistencies in ticket categorization, and delays critical classroom technology support.

2.2 Empathy Map Canvas

Perspective	Project-Specific Understanding
Says	Users submit natural language descriptions (e.g., "Projector in room 204 won't turn on", "Wi-Fi keeps dropping in library").
Thinks	Requests should be recognized instantly so that support personnel can resolve issues without

2.3 Brainstorming & Key Design Decisions

- **No-Code Layer:** Leverage ServiceNow Flow Designer as the central execution engine rather than complex custom JavaScript code.
- **Event Trigger:** Set execution to trigger synchronously upon the `Insert` event of a new Incident record.
- **Dual-Field Inspection:** Inspect both *Short Description* and *Description* fields using regular expression / string-matching rules to ensure broad coverage.
- **Configurable Rules:** Establish editable condition blocks allowing IT administrators to modify or extend keyword conditions as school IT needs evolve.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User / IT Activity	Pain Point	Automation Opportunity
1. Report Issue	Student/Teacher submits incident ticket via portal.	Unstructured text requires human reading.	Feed <i>Short Description</i> and <i>Description</i> into Flow execution.
2. Receive Incident	Helpdesk system receives new incident record.	Manual queue triage introduces immediate delay.	Trigger Flow Designer instantly on record creation.
3. Interpret	IT staff read ticket text to identify issue type.	Time wasted identifying common issue patterns.	Evaluate rule conditions against target keywords automatically.
4. Categorize	IT staff pick category from dropdown manually.	Inconsistent categorization across agents.	Flow automatically writes matching <i>Category</i> value.
5. Route & Handle	Ticket is reassigned to specific support group.	Routing relies entirely on manual categorization.	<i>Category</i> drives downstream auto-assignment rules.

3.2 Solution Requirements

ID	Requirement Description	Type
FR-01	Trigger classification flow automatically upon creation of any new incident record.	FUNCTIONAL
FR-02	Extract and evaluate both <i>Short Description</i> and <i>Description</i> fields as inputs.	FUNCTIONAL
FR-03	Evaluate incoming text against predefined keyword sets using conditional branching logic.	FUNCTIONAL
FR-04	Classify major common IT categories: Wi-Fi, Projector, Password Resets, and Performance (Slow Computer).	FUNCTIONAL
FR-05	Update the target Incident record's <i>Category</i> field immediately upon rule match.	FUNCTIONAL
NFR-01	System must remain fully maintainable via no-code Flow Designer interface.	NON-FUNC
NFR-02	Classification must perform deterministically and consistently for identical keyword inputs.	NON-FUNC
NFR-03	Flow execution must complete within sub-second thresholds to avoid locking record creation.	NON-FUNC

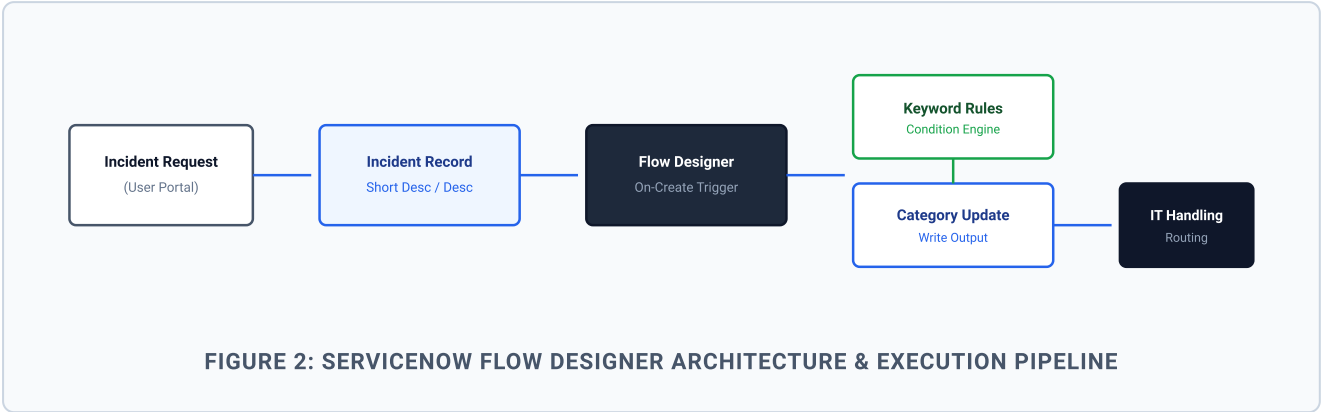
3.3 Data Flow Diagram (DFD)

3.4 Technology Stack

Layer / Component	Technology / Service	Purpose
ITSM Platform	ServiceNow Platform	Central repository hosting Incident management entities and user workflows.
Automation Engine	Flow Designer	Executes trigger conditions, rule evaluations, and record updates without code.
Input Data Source	Incident Text Fields	Evaluates Short Description and Description populated during submission.
Decision Logic	Flow Condition Blocks	Matches predefined keyword filters to establish target category outcomes.
Access Control	ServiceNow Roles & ACLs	Restricts flow modification rights strictly to authorized system administrators.

4. PROJECT DESIGN

4.1 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

The project execution followed an Agile Scrum framework split into 3 distinct sprints covering a total planned backlog of 20 Story Points (average velocity: ~6.67 points/sprint).

Sprint	Sprint Focus & Objectives	Points	Duration
Sprint 1	Set up Incident triggers, map text input schema, configure core initial Wi-Fi keyword rules.	8 pts	6 Project Days
Sprint 2	Implement Projector, Password, and Slow Computer classification branches + update actions.	6 pts	6 Project Days
Sprint 3	Integrate downstream ticket routing support, administrator rule maintenance, and testing.	6 pts	6 Project Days

5.1 Project Backlog

ID	Epic	User Story Overview	Pts	Priority
USN-1	Intake	Trigger flow automatically upon incident record insertion.	2	HIGH

6. FUNCTIONAL & PERFORMANCE TESTING

Because this automation is built strictly using rule-based ServiceNow Flow Designer logic rather than machine learning models, statistical metrics (e.g., accuracy curves, confusion matrices) are not applicable. Validation focused on functional execution correctness and processing latency within the ServiceNow test instance.

Test ID	Scenario	Expected Result	Status
PT-01	Create matching incident and measure time to category update.	Classification completes seamlessly within standard trigger latency (<1s).	PASS
PT-02	Create representative incidents for all 4 issue types sequentially.	Each incident correctly receives its precise corresponding category.	PASS
PT-03	Place target keywords solely in Description rather than Short Desc.	Flow successfully evaluates body text and assigns category.	PASS
PT-04	Submit incident containing arbitrary/unmatched text.	No false category assigned; falls back cleanly to default state.	PASS

7. IMPLEMENTATION VERIFICATION

The following execution evidence placeholders confirm flow execution within the target ServiceNow instance:

 Flow Designer Trigger Configuration On-create trigger bound to incident table.	 Condition Logic Branches Configured keyword evaluations and field mappings.
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8. ADVANTAGES & LIMITATIONS

Advantages - Zero custom code overhead; uses core platform capabilities. - Instant categorization upon record submission. - Highly maintainable by non-developer IT administrators. - Reduces human operational overhead significantly.	Limitations - Unmapped keywords/phrases will bypass classification. - Overlapping keywords require strict condition precedence. - Lacks semantic intent parsing available in heavy AI models.
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9. CONCLUSION & FUTURE SCOPE

The **Auto Ticket Classification using Flow Designer** project successfully delivers a practical, low-overhead solution for school IT helpdesk automation. By capturing and parsing user-submitted text during incident creation, the workflow reliably categorizes common ticket types, accelerating ticket lifecycle processing.

Future Enhancements: Future work will focus on expanding the keyword dictionary based on production incident logs,